

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester of B. Tech (EE) Examination****April 2018****EE-204/EE-204.01 Electrical Power System-I****Date: 30/04/2018, Monday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Define the following terms. [07]**

- i. Connected load
- ii. Maximum demand
- iii. Demand factor
- iv. Average load
- v. Load factor
- vi. Diversity factor.
- vii. Plant capacity factor.

Q – 2 Answer the questions below.

- a. Write down the properties of insulating material for cables. [07]
- b. Draw and explain Single line diagram of a typical A.C. power supply scheme. [07]

OR**Q – 2 Answer the questions below.**

- a. A 33 kV single core cable has a conductor diameter of 1 cm and a sheath of inside diameter 4 cm. find the maximum and minimum stress in the insulation. [07]
- b. Define string efficiency. Explain any two methods of improving string efficiency. [07]

Q – 3 Answer the questions below.(Any two)

- a. A diesel station supplies the following loads to various consumers: [07]
 Industrial consumer = 1500 kW ; Commercial establishment = 750 kW
 Domestic power = 100 kW ; Domestic light = 450 kW
 If the maximum demand on the station is 2500 kW and the number of kWh generated per year is 45×10^5 , determine (i) diversity factor (ii) annual load factor.
- b. Define tariff and explain different types of tariff in detail. [07]

- c. Define depreciation and explain any one method of determining depreciation. [07]

OR

Q – 3 Answer the questions below. (Any two)

- a. Define power factor. Explain disadvantages of low power factor. Also give causes of low power factor. [07]
- b. Draw and Explain the capacitance of single core cable. [07]
- c. In a 33 kV overhead line, there are three units in the string of insulators. If the capacitance between each insulator pin and earth is 11% of self-capacitance of each insulator, find (i) the distribution of voltage over 3 insulators and (ii) string efficiency. [07]

SECTION – II

Q - 4 Answer the questions below.

- a. What is skin effect? Why it is absent in the dc system? [02]
- b. Give the classification of overhead transmission lines. [03]
- c. Define voltage regulation and transmission efficiency. [02]

Q – 5 Answer the questions below.

- a. Derive an expression for the capacitance of a 3-phase overhead line for symmetrical spacing. [07]
- b. Derive an expression for the loop inductance of a single phase line in terms of GMR. [07]

OR

Q – 5 Answer the questions below.

- a. Derive an expression for the flux linkage due to internal flux of single current carrying conductor. [07]
- b. Explain Nominal π method for medium transmission line with its phasor diagram. [07]

Q – 6 Answer the questions below.

- a. Derive the model of short transmission line with phasor diagram. [07]
- b. Explain Nominal T method for medium transmission line with its phasor diagram. [07]

OR

Q – 6 Answer the questions below.

- a. Estimate the distance over which a load of 15000 kW at a p.f 0.8 lagging can be delivered by a 3-phase transmission line having conductors each of resistance 1Ω per [07]

kilometer. The voltage at the receiving end is to be 132 kV and the loss in the transmission is to be 5%.

- b. Calculate the capacitance of a 100 km long, 50 Hz overhead transmission line **[07]** consisting of 3 conductors, each of diameter 2 cm and spaced 2.5 m at the corners of an equilateral triangle.